Teacher Guide & Answers

Passage Reading Level: Lexile 1090

1. According to the passage, which gas is generated in especially large amounts when we burn fossil fuels?

- **A** **carbon dioxide**
- **B** methane
- **C** carbon monoxide
- **D** sulfur dioxide

2. The increase of carbon dioxide in the atmosphere has had which effect on the Earth's climate?

- **A** Average rainfall has decreased.
- **B** Extreme weather has become less common.
- **C** **The Earth's temperatures are rising.**
- **D** The Earth's temperatures are falling.

3. Arctic ice plays an important role in cooling the Earth. Which evidence from the passage best supports this statement?

- **A** In the last 30 years, more than one-third of the ice that appears in the Arctic during the summer has melted away.
- **B** **Arctic ice reflects the Sun's light.**
- **C** When Arctic ice melts, it reveals more of the darker-colored land or ocean water.
- **D** Darker-colored land absorbs the Sun's light, causing the climate's temperature to increase.

4. Some of the gases that contribute to global warming are trapped in permafrost. When permafrost melts, many of these gases are released into the atmosphere. This leads to an increase of the atmosphere's temperature, which causes more permafrost to melt.

What type of feedback loop is this an example of?

- **A** negative feedback loop
- **B** complex feedback loop
- **C** both a positive and negative feedback loop
- **D** **positive feedback loop**

5. What is the main idea of this passage?

- **A** **Climate change is a complex and unpredictable process involving feedback loops.**
- **B** Disagreements about climate change have prevented scientists from finding real solutions to global warming.
- **C** Wallace Broecker's theory may have been incorrect, but he presented some worthwhile ideas.
- **D** Climate change can be completely reversed if carbon dioxide production is changed.

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6. Read the following sentences: "Some scientists expect an increase in so-called 'extreme weather' events, such as hurricane and floods. Others **foresee** a rise in levels of sea water."

Which word could best replace "**foresee**" as used in this sentence?

A **forecast**
B glimpse
C pretend
D discover

7. Choose the answer that best completes the sentence below.

Trees play important roles in preventing global warming, _____ absorbing carbon dioxide and pumping water into the atmosphere.

A thus
B finally
C as a result
D **including**

8. How does carbon dioxide increase the Earth's temperature?

Suggested answer: Carbon dioxide traps heat which can cause the Earth's temperature to rise if large quantities of carbon dioxide are released in the Earth's atmosphere.

9. What is a feedback loop?

Suggested answer: A feedback loop is a process that helps to amplify or diminish certain changes within the process.

10. How can heat produce more heat? Use information from the passage to support your answer.

Suggested answer: Answers may vary and should be supported by the passage. Students should indicate that heat produces more heat in positive feedback loop systems. Students may also illustrate this phenomenon using one of the examples from the passage. For example, they may mention the example of melting permafrost. Some of the gases that contribute to global warming are trapped in permafrost. When permafrost melts, many of these gases are released into the atmosphere. This leads to an increase of the atmosphere's temperature, which causes more permafrost to melt. As more permafrost melts, the atmosphere's temperature increases. Thus, heat can produce more heat.

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